Wotmed-Net (ወጥመድ-ኔት)

INSA Cyber Talent Summer Camp | Group 1 Master Proposal Breakdown & Role Specifications

This document details the precise, deep-dive proposal documentation and architectural responsibilities assigned to each member of Group 1. To ensure a professional execution pipeline starting from zero, each member is responsible for drafting, researching, and finalizing their specialized technical domain before collation into the master project proposal.

1. Ayitegeb Tilahun Demisie

Lead Architect & Integration

Core Middleware Engine & System Integration Proposal

Primary Objective: Formulate the overarching system architecture, define cross-component communication protocols, and establish the main middleware engine pipeline.

- **System Architecture Overview:** Documenting the multi-layered design of Wotmed-Net, detailing how requests flow from incoming perimeter traffic through the active defense filter and into protected backend applications.
- **Core Middleware Engine Specifications:** Defining the interception logic, packet inspection hooks, rule evaluation engine, and execution lifecycle within the middleware environment.
- **Integration & API Contracts:** Establishing standardized JSON/REST and WebSocket payload schemas for seamless communication between the middleware core, telemetry sink, and the React SOC dashboard.
- **Failure Resilience & Performance Bounds:** Specifying latency thresholds, fail-safe bypass mechanisms to ensure application availability under heavy traffic or engine failure, and resource isolation protocols.

> **Deliverable Document:** `/docs/proposal/01-architecture-and-core-engine.md`

2. Dilargachew Ashenafi Tsegemariam

Backend & Real-Time Sync

Telemetry Ingestion & WebSocket Pipelines Proposal

Primary Objective: Design the robust backend infrastructure capable of ingesting high-throughput security telemetry and streaming real-time alerts to the monitoring interface.

- **Backend Infrastructure Architecture:** Designing the server-side environment (Node.js/Express or Python FastAPI), database schema design (Supabase/PostgreSQL), and connection pooling.
- **Telemetry Ingestion Pipeline:** Creating high-performance event intake endpoints designed to receive structured security logs, threat payloads, and heuristic violations from the middleware engine with minimal latency.
- **Real-Time Synchronization (WebSockets):** Implementing bi-directional WebSocket communication channels and pub/sub event emitters to broadcast threat alerts instantly to connected dashboard clients.
- **Data Retention & Storage Optimization:** Defining indexing strategies, log rotation, and historical data querying for incident analysis and telemetry auditing.

> **Deliverable Document:** `/docs/proposal/02-backend-and-telemetry.md`

3. Biruk Zelalem Yohannes

SOC Dashboard Lead

Dark-Mode React Threat Operations UI Proposal

Primary Objective: Engineer the front-end Security Operations Center (SOC) dashboard specifications, user experience workflows, and real-time visualization components.

- **UI/UX Design System & Theme:** Establishing a specialized tactical dark-mode aesthetic optimized for low-fatigue monitoring during incident response and threat tracking.
- **Component Hierarchy & State Management:** Structuring the React/Next.js component tree (live threat feeds, IP geolocation mapping, alert severity matrices, telemetry charts) using modern state management libraries.
- **Real-Time Data Binding:** Designing robust WebSocket event listeners and client-side state updaters to reflect incoming security telemetry without requiring page refreshes.
- **Interactive Incident Management:** Outlining UI controls for security analysts to block malicious IPs, inspect captured payloads, adjust honeytoken sensitivity, and export incident reports.

> **Deliverable Document:** `/docs/proposal/03-soc-dashboard-ui.md`

4. Eman Mohammednur Ahmed

Demo App & Deception Tester

Reference App & Honeytoken Configurations Proposal

Primary Objective: Develop the reference vulnerable application environment and establish deceptive honeytoken deployment strategies to lure and trap attackers.

- **Reference Application Architecture:** Building a standard multi-tier sample web application (e.g., simulated enterprise portal or e-commerce shop) designed to integrate directly with the Wotmed-Net middleware.
- **Honeytoken & Canary Design:** Strategically embedding hidden form fields, invisible administrative endpoints, fake database credentials, and canary files designed exclusively to detect unauthorized probing.
- **Deception Workflow & Trigger Mapping:** Documenting how interaction with honeytokens immediately generates high-priority telemetry alerts and triggers active defense countermeasures.
- **Verification Testing Scenarios:** Creating baseline functional tests to ensure the reference application remains fully operational for legitimate users while deceptive traps remain active.

> **Deliverable Document:** `/docs/proposal/04-demo-app-and-deception.md`

5. Wubalem Gashaw Abera

Red Team & Security Tester

Automated Exploit Probing & Presentation Lead

Primary Objective: Establish automated testing frameworks, simulate sophisticated attack vectors to validate defense efficacy, and lead the final project presentation structure.

- **Automated Exploit Probing Framework:** Designing script-based attack simulators (Python/Bash) targeting SQL injection, cross-site scripting (XSS), path traversal, and brute-force authentication.
- **Defense Validation & Efficacy Testing:** Executing systematic penetration tests against Wotmed-Net to measure detection rates, false positive ratios, and response latencies under simulated adversarial pressure.
- **Red Team Reporting Metrics:** Formulating standardized evaluation criteria to score how effectively the active defense middleware blocks attacks and feeds telemetry to the SOC dashboard.
- **Final Presentation & Documentation Lead:** Structuring the comprehensive slide deck, executive summary, and live demonstration script for the final project defense.

> **Deliverable Document:** `/docs/proposal/05-security-testing-and-presentation.md`